

Sprint 3

Stroke Prediction Analytics using Python

Advanced Statistics & Feature Engineering

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Batch: 544-A

Dataset: healthcare-dataset-stroke-data.csv

Sprint 3 used the cleaned 5,110-record dataset and the findings from the earlier exploratory analysis. The purpose of this sprint was to check important relationships statistically, calculate confidence intervals, create useful features and turn the results into practical recommendations.

1. Statistical Analysis

The notebook calculated central tendency, dispersion, distributions, correlation and covariance for the numerical variables.

Variable	Mean	Median	Standard deviation
Age	43.2266	45.0000	22.6126
Average glucose level	106.1477	91.8850	45.2836
BMI	28.8620	28.1000	7.6996
Hypertension	0.0975	0	0.2966
Heart disease	0.0540	0	0.2261
Stroke	0.0487	0	0.2153

Age had a small negative skew of -0.137, while average glucose level had positive skew of 1.572 and BMI had positive skew of 1.088. This agrees with the earlier observation that glucose and BMI have longer right tails.

1.1 Correlation

Variable	Correlation with stroke
Age	0.245257
Heart disease	0.134914
Average glucose level	0.131945
Hypertension	0.127904
BMI	0.036110

Age again had the strongest positive numerical relationship with stroke. Heart disease, glucose level and hypertension also had positive relationships, while BMI had a much weaker relationship.

1.2 Covariance

The covariance matrix showed positive covariance between several health variables. For example, age and average glucose level had positive covariance, and age also had positive covariance with stroke. Covariance helps show direction, while correlation is easier to compare because it is standardized.

2. Hypothesis Testing

Three statistical tests were used in the notebook. A significance level of 0.05 was used for the decisions.

2.1 T-Test: Age and Stroke

Null hypothesis: the mean age of stroke and non-stroke patients is the same. Alternate hypothesis: the mean ages are different.

Result	Value
T-statistic	29.6863
p-value	2.1157×10^{-95}
Decision	Reject the null hypothesis

The very small p-value provides strong statistical evidence that average age differs between stroke and non-stroke patients. This supports the earlier exploratory finding that stroke patients in this dataset are older on average.

2.2 Chi-Square Test: Hypertension and Stroke

Null hypothesis: there is no association between hypertension and stroke. Alternate hypothesis: there is an association between hypertension and stroke.

Result	Value
Chi-square statistic	81.6054
p-value	1.6616×10^{-19}
Degrees of freedom	1
Decision	Reject the null hypothesis

The result gives strong statistical evidence of an association between hypertension and stroke in this dataset. It does not by itself prove that hypertension causes stroke.

2.3 ANOVA: Age and Work Type

Null hypothesis: the mean age is the same across all work-type groups. Alternate hypothesis: at least one work-type group has a different mean age.

Result	Value
F-statistic	1110.0851
p-value	0.0 as returned by the notebook

Decision	Reject the null hypothesis
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The ANOVA result indicates a statistically significant difference in average age across work-type groups. This test is about age differences between work groups, not about proving that work type causes stroke.

3. Confidence Interval Analysis

The notebook calculated 95% confidence intervals for age, average glucose level and BMI.

Feature	Mean	95% confidence interval
Age	43.2266	42.6065 to 43.8468
Average glucose level	106.1477	104.9058 to 107.3896
BMI	28.8620	28.6509 to 29.0732

These intervals provide plausible ranges for the corresponding population means represented by the sample. They also show the uncertainty around the sample mean instead of giving a single number only.

4. Feature Engineering

The engineered features were created using findings from the previous sprint and were intended to make the patient information easier to compare.

New feature	Purpose
age_group	Groups age into Young, Adult, Middle_Age and Senior.
bmi_category	Groups BMI into Underweight, Normal, Overweight and Obese.
combined_disease_indicator	Adds hypertension and heart disease indicators.
disease_status	Labels patients as No Disease, One Condition or Both Conditions.
glucose_category	Groups glucose into Low, Normal, High and Very_High.
health_score	Simple count of hypertension and heart disease indicators.

Age Group was created because age was an important variable in the earlier analysis. BMI Category makes the continuous BMI value easier to compare. The combined disease features summarize hypertension and heart disease together. Glucose Category makes the glucose distribution easier to interpret. The Health Score is a simple project-specific analytical measure and is not a clinical score.

5. Feature Evaluation

The new features were evaluated using patient counts, stroke cases and stroke rates.

5.1 Age Group

Age group	Patients	Stroke rate
Young	856	0.23%
Adult	1,314	0.46%
Middle_Age	1,564	3.84%
Senior	1,376	13.15%

The age groups make the age pattern much easier to see. The observed stroke rate rises strongly across the groups, with the Senior group having the highest rate.

5.2 BMI Category

BMI category	Patients	Stroke rate
Underweight	349	0.29%
Normal	1,258	2.94%
Overweight	1,610	7.14%
Obese	1,893	5.07%

BMI categories provide a simple way to compare groups. In this dataset the Overweight group had the highest observed stroke rate, followed by the Obese group. This is an observed pattern and should not be treated as proof of causation.

5.3 Disease Status

Disease status	Patients	Stroke rate
No Disease	4,400	3.39%
One Condition	646	13.47%
Both Conditions	64	20.31%

Disease Status gives a useful combined view of hypertension and heart disease. The observed stroke rate increases from patients with no condition to patients with one condition and then to patients with both conditions.

5.4 Glucose Category

Glucose category	Patients	Stroke rate
Low	756	3.57%

Normal	2,375	3.58%
High	998	3.71%
Very High	981	10.19%

The glucose grouping makes the higher stroke rate in the Very High category easier to identify. The other categories have similar observed rates in comparison.

6. Final Business Recommendations

- Age should receive attention in stroke-risk assessment because older groups had much higher observed stroke rates.
- Hypertension and heart disease should be considered when reviewing patient health profiles because both were associated with higher observed stroke rates.
- Very high glucose levels can be included as one of the indicators reviewed during preventive-care analysis.
- The combined disease status is useful for identifying patients with one or multiple health conditions.
- The engineered age, BMI, disease and glucose groups make the findings easier to communicate and compare.
- The stroke class imbalance should be considered in any future predictive modelling.
- Statistical association should not be interpreted as direct causation.
- These findings are analytical observations from the dataset and are not a substitute for clinical judgement or medical diagnosis.

7. Sprint 3 Conclusion

Sprint 3 strengthened the findings from the earlier exploratory analysis by using statistical tests and confidence intervals. The hypothesis tests showed statistically significant differences or associations for the relationships tested. The engineered features also made the patient groups easier to compare, especially age groups and combined disease status. The final recommendations connect these findings to preventive-care analysis while keeping the limitations of the dataset in mind.

Final Sprint 3 dataset: 5,110 rows and 18 columns after feature engineering.

Source notebook: Sprint_3_based_Ap.ipynb